

Minjae Seo

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EDUCATION

University of Chicago, Harris School of Public Policy

June 2027

Master of Arts in Public Policy, Certificate in Research Methods

GPA: 3.7/4.0

Coursework: Applied Econometrics I–III (PhD), Advanced Microeconomics for Policy Analysis I–II (PhD), Game Theory (PhD), Bayesian, AI and Deep Learning (PhD), Analysis in R^n I–II, Introduction to Stochastic Processes, Economic Development and Policy, Research Methodology and Writing

University of California, Berkeley

Dec 2024

Bachelor of Arts in Statistics

GPA: 3.85/4.0

Coursework: Modern Statistical Prediction & Machine Learning, Principles & Techniques of Data Science, Econometrics, Causal Inference, Optimization Models in Engineering, Game Theory

TECHNICAL SKILLS

Languages: Python, SQL, R, Stata, C, MATLAB, SAS

Toolkit: pandas, NumPy, scikit-learn, matplotlib, fixest, rdrobust, dplyr, ggplot2, Git, Quarto, LaTeX, AWS, WRDS Cloud, Tableau, Power BI, React, ETL

EXPERIENCE

Research Analyst Intern, World Bank, Washington, DC

Jun 2026 – Present

- Analyzed two nationwide multi-arm A/B experiments for Pakistan's Federal Board of Revenue (88,229 companies across 8 message arms; 102,130 engineers across 6), where messaging raised filing by 2.06pp and 3.96pp but over 90% of the corporate filings it caused were empty.
- Established that only the engineer campaign moved revenue (+1.3pp positive liability, +PKR 455 per taxpayer), redirecting program reporting from filing counts to declared liability.
- Produced a 17-artifact reproducible R pipeline across five commissioned analyses, resolving 27 reviewer points and pulling a headline +81% figure that conditioned on a post-treatment outcome.
- Built the Lumina evidence dashboard on 30 verified youth-employment RCTs, rejecting one of 18 supplied references as not an RCT.

Research Assistant, Booth School of Business, University of Chicago

Jan 2026 – Jun 2026

- Delivered a 2.38B-row (4.90 TiB) U.S. job-postings dataset on WRDS Cloud using chunked state-by-year SQL with checkpoint/resume, so the extraction runs unattended and restarts from the last completed chunk.
- Consolidated 899 Excel sources (448,119 citation rows, 1920–2000) into a ranked 14,500-journal database, using fuzzy name matching, three-pass QC, and a parallel Python build so the ranking reconciles across two independent implementations.
- Mapped the library's access coverage by classifying all 14,500 journals across four digital platforms.

AI Intern, YCX, Remote

Mar 2025 – May 2025

- Shipped an AI know-your-business compliance system to production, automating regulatory due-diligence review.
- Grew the companies the system can screen by 40%, to 300+, with a modular pipeline of LLM agents (Groq API) and targeted web scraping.

Quantitative Research Intern, Global Key Advisors

Jul 2024 – Mar 2025

- Engineered reusable Python extraction and ROI-tracking pipelines that fed the team's downstream strategy backtests, merging SEC Form 4, CRSP, and Refinitiv across the Fama–French 48 industries.
- Tested whether insider-disclosure rules leave exploitable alpha using event-time abnormal returns, benchmarked against Fama–French factors.

PROJECTS

Community-College Transfer Earnings Gap (R, NLSY97 microdata)

Sep 2025 – May 2026

- Measured the earnings gap between community-college transfer students and four-year entrants: \$7,770 less per year ($p = 0.048$, $N = 2,306$).
- Confirmed the penalty across five estimators (OLS, propensity matching, IPW, AIPW), bracketing it between \$7,770 and \$5,189, in line with a published administrative benchmark.
- Showed a widely cited explanation was a statistical artifact: holding the comparison group fixed shrank it from 27.5% to 4.5%.

Retail Credit Default Prediction (Python, scikit-learn)

Mar 2026 – Apr 2026

- Trained a default-risk model on a loan book with a 5.4% default rate and 100+ correlated features, using L1 selection, SMOTE, and cost-sensitive learning, and scoring on F1/AUCPR so the metric tracks caught defaults rather than accuracy.
- Wrote a concept-drift detection and retraining policy so post-deployment decay is caught on a schedule, with notebooks published.

AI Exposure and Labor Demand (Python, Revelio Labs + Compustat)

Mar 2026 – Jun 2026

- Assembled a 14.5M-cell panel linking Revelio Labs job postings to Compustat financials (2018–2025), which supported a triple-difference across occupation AI exposure, firm size, and the pre/post-ChatGPT split.
- Estimated sharp declines in AI-exposed postings at small (-0.211 , SE 0.033) and large (-0.146 , SE 0.026) firms, but reported the gap as a null ($+0.065$, $p = 0.14$).

TEACHING & LEADERSHIP

Teaching Assistant, Microeconomics (MBA core), Booth School of Business, University of Chicago

Sep 2026 – Present

- Run the weekly review session for 180 MBA students across seven graded problem sets and teach the opening calculus refresher.

Graduate Teaching Assistant, CMSC 35300 Mathematical Foundations of ML, Dept. of Computer Science, University of Chicago

Sep 2025 – Dec 2025

- Taught weekly recitations for 20+ master's students, translating SVD, Gram–Schmidt, and least squares into Python/NumPy demos.
- Answered 50+ technical questions on Ed Discussion.

Head of Research, Undergraduate Economics Association, UC Berkeley

Jan 2024 – Mar 2025

- Led the research program of a 30+ member organization, running biweekly workshops in regression (OLS, IV, DiD, RDD) with R/Stata.
- Mentored members through semester-long projects, from identification strategy to written deliverables.

Competitor, Econometrics Game, University of Chicago

Mar 2024

- Selected as one of four (two LSE, one UC Berkeley) to produce a complete empirical study in 48 hours; presented the team's final analysis to a panel including a Nobel Laureate.